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Cohort: Cohort C

CREATE DATABASE projectsschema;

USE projectsschema;

Database changed

====QUESTION 1=====

CREATE TABLE EMPLOYEE (

-> EMPNO VARCHAR(10) PRIMARY KEY,

-> FNAME VARCHAR(30),

-> LNAME VARCHAR(40),

-> ADDRESS VARCHAR(20),

-> DOB DATE,

-> SEX CHAR(1), CONSTRAINT CHK_SEX CHECK (SEX IN ('M', 'F')),

-> POSITION VARCHAR(20), CONSTRAINT CHK_POSITION CHECK (POSITION IN ('Manager', 'Team Leader', 'Analyst', 'Software Developer')),

-> DEPTNO VARCHAR(10));

DESCRIBE EMPLOYEE;

Field	Type	Null	Key	Default	Extra
EMPNO	varchar(10)	NO	PRI	NULL	
FNAME	varchar(30)	YES		NULL	
LNAME	varchar(40)	YES		NULL	
ADDRESS	varchar(20)	YES		NULL	
DOB	date	YES		NULL	
SEX	char(1)	YES		NULL	
POSITION	varchar(20)	YES		NULL	
DEPTNO	varchar(10)	YES		NULL	

CREATE TABLE DEPARTMENT (

-> DEPTNO VARCHAR(10),

-> DEPTNAME VARCHAR(30),

-> MGREMPNO VARCHAR(10),

-> PRIMARY KEY (DEPTNO),

-> FOREIGN KEY (MGREMPNO) REFERENCES EMPLOYEE(EMPNO)

->);

DESCRIBE DEPARTMENT;

Field	Type	Null	Key	Default	Extra
DEPTNO	varchar(10)	NO	PRI	NULL	

DEPTNAME	varchar(30)	YES	NULL	
MGREMPNO	varchar(10)	YES	MUL	NULL

+-----+-----+-----+-----+-----+

```
CREATE TABLE PROJECT (
    -> PROJNO VARCHAR(10),
    -> PROJNAME VARCHAR(30),
    -> DEPTNO VARCHAR(10),
    -> PRIMARY KEY(PROJNO),
    -> FOREIGN KEY (DEPTNO) REFERENCES DEPARTMENT(DEPTNO)
    -> );
```

```
DESCRIBE PROJECT;
```

+-----+	+-----+	+-----+	+-----+	+-----+	+-----+
Field	Type	Null	Key	Default	Extra
+-----+	+-----+	+-----+	+-----+	+-----+	+-----+
PROJNO	varchar(10)	NO	PRI	NULL	
PROJNAME	varchar(30)	YES		NULL	
DEPTNO	varchar(10)	YES	MUL	NULL	
+-----+	+-----+	+-----+	+-----+	+-----+	+-----+

```
CREATE TABLE WORKSON (
    -> EMPNO VARCHAR(10),
    -> PROJNO VARCHAR(10),
```

-> DATEWORKED DATE,
 -> HOURSWORKED INT,
 -> PRIMARY KEY (EMPNO, PROJNO, DATEWORKED),
 -> FOREIGN KEY (PROJNO) REFERENCES PROJECT(PROJNO),
 -> FOREIGN KEY (EMPNO) REFERENCES EMPLOYEE(EMPNO),
 -> CONSTRAINT CHK_HOURS CHECK (HOURSWORKED BETWEEN 0 AND 40)
 ->);

DESCRIBE WORKSON;

Field	Type	Null	Key	Default	Extra
EMPNO	varchar(10)	NO	PRI	NULL	
PROJNO	varchar(10)	NO	PRI	NULL	
DATEWORKED	date	NO	PRI	NULL	
HOURSWORKED	int(11)	YES		NULL	

ALTER TABLE EMPLOYEE

-> ADD FOREIGN KEY (DEPTNO) REFERENCES DEPARTMENT(DEPTNO);

DESCRIBE EMPLOYEE;

Field	Type	Null	Key	Default	Extra
-------	------	------	-----	---------	-------

```

| EMPNO | varchar(10) | NO | PRI | NULL | |
| FNAME | varchar(30) | YES | | NULL | |
| LNAME | varchar(40) | YES | | NULL | |
| ADDRESS | varchar(20) | YES | | NULL | |
| DOB | date | YES | | NULL | |
| SEX | char(1) | YES | | NULL | |
| POSITION | varchar(20) | YES | | NULL | |
| DEPTNO | varchar(10) | YES | MUL | NULL | |
+-----+-----+-----+-----+-----+-----+

```

INSERT INTO EMPLOYEE (EMPNO, FNAME, LNAME, ADDRESS, DOB, SEX, POSITION, DEPTNO) VALUES

```

-> ('E1', 'John', 'Smith', '12 Main St', '1985-03-12', 'M', 'Manager', NULL),
-> ('E2', 'Mary', 'Jones', '45 Oak Ave', '1990-07-22', 'F', 'Manager', NULL),
-> ('E3', 'David', 'Brown', '8 Elm Rd', '1992-11-05', 'M', 'Analyst', NULL),
-> ('E4', 'Sarah', 'Davis', '21 Pine St', '1995-01-18', 'F', 'Software Developer', NULL),
-> ('E5', 'Janice', 'Wilson', '3 Birch Ln', '1991-09-30', 'F', 'Team Leader', NULL)
;

```

SELECT * FROM EMPLOYEE;

```

+-----+-----+-----+-----+-----+-----+-----+
| EMPNO | FNAME | LNAME | ADDRESS | DOB | SEX | POSITION | DEPTNO |
+-----+-----+-----+-----+-----+-----+-----+
| E1 | John | Smith | 12 Main St | 1985-03-12 | M | Manager | D2 |
| E2 | Mary | Jones | 45 Oak Ave | 1990-07-22 | F | Manager | D1 |

```

E3	David	Brown	8 Elm Rd	1992-11-05	M	Analyst	D2
E4	Sarah	Davis	21 Pine St	1995-01-18	F	Software Developer	D1
E5	Janice	Wilson	3 Birch Ln	1991-09-30	F	Team Leader	D2

```

INSERT INTO DEPARTMENT (DEPTNO, DEPTNAME, MGREMPNO) VALUES
    -> ('D1', 'IT', 'E2'),
    -> ('D2', 'Planning', 'E1')
;

```

```

SELECT * FROM DEPARTMENT;

```

DEPTNO	DEPTNAME	MGREMPNO
D1	IT	E2
D2	Planning	E1

```

UPDATE EMPLOYEE SET DEPTNO = 'D1' WHERE EMPNO IN ('E2','E4');
UPDATE EMPLOYEE SET DEPTNO = 'D2' WHERE EMPNO IN ('E1','E3','E5');

```

```

SELECT * FROM EMPLOYEE;

```

EMPNO	FNAME	LNAME	ADDRESS	DOB	SEX	POSITION	DEPTNO
-------	-------	-------	---------	-----	-----	----------	--------

E1	John	Smith	12 Main St	1985-03-12	M	Manager	D2
E2	Mary	Jones	45 Oak Ave	1990-07-22	F	Manager	D1
E3	David	Brown	8 Elm Rd	1992-11-05	M	Analyst	D2
E4	Sarah	Davis	21 Pine St	1995-01-18	F	Software Developer	D1
E5	Janice	Wilson	3 Birch Ln	1991-09-30	F	Team Leader	D2

INSERT INTO PROJECT (PROJNO, PROJNAME, DEPTNAME) VALUES

-> ('P1', 'SCCS', 'D1'),
-> ('P2', 'Website Redesign', 'D1'),
-> ('P3', 'Resource Planning', 'D2')
;

SELECT * FROM PROJECT;

PROJNO	PROJNAME	DEPTNO
P1	SCCS	D1
P2	Website Redesign	D1
P3	Resource Planning	D2

INSERT INTO WORKSON (EMPNO, PROJNO, DATEWORKED, HOURSWORKED) VALUES

-> ('E2', 'P1', '2024-01-10', 35),

```
-> ('E4', 'P1', '2024-01-10', 20),
-> ('E4', 'P2', '2024-01-11', 15),
-> ('E1', 'P3', '2024-01-12', 40),
-> ('E3', 'P3', '2024-01-12', 30),
-> ('E5', 'P3', '2024-01-13', 25)
;
```

```
SELECT * FROM WORKSON;
```

```
+-----+-----+-----+-----+
| EMPNO | PROJNO | DATEWORKED | HOURSWORKED |
+-----+-----+-----+-----+
| E1   | P3     | 2024-01-12 | 40 |
| E2   | P1     | 2024-01-10 | 35 |
| E3   | P3     | 2024-01-12 | 30 |
| E4   | P1     | 2024-01-10 | 20 |
| E4   | P2     | 2024-01-11 | 15 |
| E5   | P3     | 2024-01-13 | 25 |
+-----+-----+-----+-----+
```

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=====QUESTION 7.25=====
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```

```
CREATE VIEW FemaleManagerLeadProjects AS
```

```

-> SELECT p.PROJNO, p.PROJNAME
-> FROM PROJECT p, DEPARTMENT d, EMPLOYEE e
-> WHERE p.DEPTNO = d.DEPTNO
-> AND d.MGREMPNO = e.EMPNO
-> AND e.SEX = 'F'
-> ORDER BY PROJNO
;

```

```
SELECT * FROM FemaleManagerLeadProjects;
```

```

+-----+-----+
| PROJNO | PROJNAME |
+-----+-----+
| P1     | SCCS     |
| P2     | Website Redesign |
+-----+-----+

```

```

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=====QUESTION 7.26=====
-----

```

```
CREATE VIEW EmployeeProjectDetails AS
```

```

-> SELECT e.EMPNO, e.FNAME, e.LNAME, p.PROJNAME, w.HOURSWORKED
-> FROM EMPLOYEE e, PROJECT p, WORKSON w
-> WHERE w.EMPNO = e.EMPNO
-> AND w.PROJNO = p.PROJNO

```

-> ;

SELECT * FROM EmployeeProjectDetails;

```
+-----+-----+-----+-----+-----+
| EMPNO | FNAME | LNAME | PROJNAME      | HOURSWORKED |
+-----+-----+-----+-----+-----+
| E2   | Mary  | Jones | SCCS          | 35 |
| E4   | Sarah | Davis | SCCS          | 20 |
| E4   | Sarah | Davis | Website Redesign | 15 |
| E1   | John  | Smith | Resource Planning | 40 |
| E3   | David | Brown | Resource Planning | 30 |
| E5   | Janice | Wilson | Resource Planning | 25 |
+-----+-----+-----+-----+-----+
```

=====QUESTION 7.27=====

=== A ===

CREATE VIEW EmpProject(empNo, projNo, totalHours)

-> AS SELECT w.EMPNO, w.PROJNO, SUM(HOURSWORKED)

-> FROM EMPLOYEE e, PROJECT p, WORKSON w

-> WHERE e.EMPNO = w.EMPNO AND p.PROJNO = w.PROJNO

-> GROUP BY w.EMPNO, w.PROJNO

-> ;

SELECT * FROM EmpProject;

empNo	projNo	totalHours
E1	P3	40
E2	P1	35
E3	P3	30
E4	P1	20
E4	P2	15
E5	P3	25

=====

=== B ===

=====

EMPTY SET SHOWN

EmpProject Only stores values like P1, P2, P3 which are the project numbers and not project names like 'SCCS'

=====

=== C ===

=====

```
SELECT COUNT(PROJNO) FROM EmpProject WHERE EMPNO = 'E1';
```

```
+-----+
| COUNT(PROJNO) |
+-----+
|      1      |
+-----+
```

```
-----
=== D ===
-----
```

```
SELECT EMPNO, totalHours FROM EmpProject GROUP BY EMPNO;
```

```
+-----+-----+
| EMPNO | totalHours |
+-----+-----+
| E1    | 40         |
| E2    | 35         |
| E3    | 30         |
| E4    | 20         |
| E5    | 25         |
+-----+-----+
```

IMPORTANT NOTICE ABOUT THIS QUERY

E4 HAS TWO ROWS IN EmpProject WHICH ARE;

1. P1 = 20

2. P2 = 15

BUT GROUPING BY EMPNO ALONE COLLAPSES OR COMBINES THEM AND MARIA-DB AUTOMATICALLY PICKS

ONE VALUE i.e. 20 FOR THE totalHours INSTEAD OF ADDING THEM BOTH, MAKING THIS RESULT A

MISLEADING ONE. THOUGH THE QUERY RUNS WITHOUT ANY ERRORS THE totalHours ISNT A MEANINGFUL

VALUE.

=====QUESTION 7.28=====

THE VIEW STORES DISTINCT PART NUMBERS WITH ANY OF THEIR CONTACT COST BEING OVER 1000(POUNDS)

MAINTENANCE FOR SUCH A VIEW IS TRICKY MAINLY BECAUSE EVEN IF WE DELETE OR UPDATE AN EXPENSIVE ROW,

IT DOESNT MEAN THE PART IS AUTOMATICALLY REMOVED, THE OTHER CONTRACTS MIGHT STILL KEEP IT.

BASE TABLE ACCESS IS NOT NEEDED IN THESE INSTANCES

1. INSERTION OF AN EXPENSIVE ROW , YOU CAN JUST ADD PARTNO

2. UPDATING A CHEAP ROW TO AN EXPENSIVE ONE, YOU CAN ALSO JUST ADD PARTNO

THIS IS POSSIBLE BECAUSE THERES NO NEED FOR THE OPERATIONS TO CHECK IF OTHER ROWS EXIST

BASE TABLE ACCESS IS NEEDED IN THESE INSTANCES

1. THE DELETION OF AN EXPENSIVE ROW
2. UPDATING OF AN EXPENSIVE ROW TO A CHEAP ONE

THIS IS BECAUSE IT MUST CHECK IF ANY OTHER ROW FOR THAT PARTNO STILL HAS A COST OF OVER 1000(POUNDS),

BASICALLY IT NEEDS TO QUERY THE BASE TABLE

WE CAN AVOID BASE TABLE ACCESS BY:

1. STORING AGGREGATE COUNTS IN THE VIEW INSTEAD OF JUST THE DISTINCT PART NUMBERS
2. AND THEN MAINTAIN THE VIEW BY INCREMENTING OR DECREMENTING COUNTS USING ONLY THE

CHANGED ROW WHICH REQUIRES NO BASE TABLE READING

IN CONCLUSION

BASE TABLE IS AVOIDABLE FOR INSERTS AND COST INCREASES

BUT NOT FOR DELETES OR COST DECREASES, UNLESS THERE ARE ADDITIONAL METADAT STORED IN THE MATERIALISED

VIEW ITSELF.